Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A metal rod of Young's modulus $2 \times 10^{10}\ \mathrm{N\ m^{-2}}$ undergoes an elastic strain of 0.06%. The energy per unit volume stored in $\mathrm{Jm^{-3}}$ is

(A) 3600

(B) 7200

(C) 10800

(D) 14400

Q.2

An elastic string of unstretched length L and force constant k is stretched by a small length x. It is further stretched by another small length y. The work done in the second stretching is:

(A) $\frac{1}{2}ky^2$

(B) $\frac{1}{2}k\left(x^2+y^2\right)$

(C) $\frac{1}{2}k(x+y)^2$

(D) $\frac{1}{2}ky(2x+y)$

Q.3

The force exerted by a special compression device is given as function of compression x as $F_x(x) = kx(x-\ell)$ for $0 \le x \le \ell$, where ℓ is maximum possible compression and k is a constant. The force exerted by the device under compression is maximum when compression is

(A) 0

(B) $\ell/4$

(C) $\ell/\sqrt{2}$

(D) $\ell/2$

Q.4

The upper end of a wire of diameter $12\ \mathrm{mm}$ and length $1\ \mathrm{m}$ is clamped and its other end is twisted through an angle of 30°. The angle of shear is

(A) 18°

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(B) 0.18°

(C) 36°

(D) 0.36°

Q.5

Steel ruptures when a shear of $3.5 \times 10^{8}\ \mathrm{N\ m^{-2}}$ is applied. What amount of force (in newton) is needed to punch a 1 cm diameter hole in a steel sheet $0.3\ \mathrm{cm}$ thick nearly?

(A) 3300

(B) 330

(C) 33

(D) 33000

Q.6

A circular tube of mean radius $8\ \mathrm{cm}$ and thickness $0.04\ \mathrm{cm}$ is melted up and recast into a solid rod of the same length. The ratio of the torsional rigidities of the circular tube and the solid rod is

(A) $\frac{(8.02)^4-(7.98)^4}{(0.8)^4}$

(B) $\frac{(8.02)^2-(7.98)^2}{(0.8)^2}$

(C) $\frac{(0.8)^2}{(8.02)^4-(7.98)^4}$

(D) $\frac{(0.8)^2}{(8.02)^3-(7.98)^2}$

Q.7

In materials like aluminium and copper, the correct order of magnitude of various elastic modului is:

(A) Young's modulus $<$ shear modulus $<$ bulk modulus.

(B) Bulk modulus $<$ shear modulus $<$ Young's modulus

(C) Shear modulus $<$ Young's modulus $<$ bulk modulus.

(D) Bulk modulus $<$ Young's modulus $<$ shear modulus.

Q.8

A cube at temperature $0^{\circ}\mathrm{C}$ is compressed equally from all sides by an external pressure P. By what amount should its temperature be raised to bring it back to the size it had before the external pressure was applied. The bulk modulus of the material of the cube is B and the coefficient of linear expansion is α.

(A) $\mathrm{P}/\mathrm{B}\alpha$

(B) $\mathrm{P}/3\ \mathrm{B}\alpha$

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(C) $3\pi\alpha/\mathrm{B}$

(D) $3\ \mathrm{B}/\mathrm{P}$

Q.9

For a constant hydraulic stress on an object, the fractional change in the object volume $\left(\frac{\Delta V}{V}\right)$ and its bulk modulus
(B) are related as

(A) $\frac{\Delta V}{V} \propto B$

(B) $\frac{\Delta V}{V} \propto \frac{1}{B}$

(C) $\frac{\Delta V}{V} \propto B^2$

(D) $\frac{\Delta V}{V} \propto B^{-2}$

Q.10

Two rods A and B of the same material and length have their radii r_1 and r_2 respectively. When they are rigidly fixed at one end and twisted by the same couple applied at the other end, the ratio $\left(\frac{\text{Angle of twist at the end of A}}{\text{Angle of twist at the end of B}}\right)$ is

(A) $\mathrm{r}_1^2/\mathrm{r}_2^2$

(B) $\mathrm{r}_1^3/\mathrm{r}_2^3$

(C) $\mathrm{r}_2^4/\mathrm{r}_1^4$

(D) $\mathrm{r}_1^4/\mathrm{r}_2^4$

Q.11

The volume of a wire remains unchanged when the wire is subjected to a certain tension. The Poisson's ratio of the material of the wire is

(A) 0.25

(B) 0.3

(C) 0.4

(D) 0.5

Q.12

The Poisson's ratio of a material is 0.4. If a force is applied to a wire of this material, there is a decrease of cross-sectional area by 2%. The percentage increase in its length is:

(A) 3%

(B) 2.5%

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(C) 1%

(D) 0.5%

Q.13

Consider a 1 m long steel wire with the cross-section of 0.05 cm^2 and $Y_{steel} = 2 \times 10^{11} Nm^{-2}$. Now, a 15 kg mass is attached to the end of the wire and it is whirled in a vertical circle with an angular frequency of 2 rev / s at the bottom of the circle. When the mass is at the lowest point of its path, find the elongation of the wire in mm: (Take $g = 10$ ms^{-2} & $\pi^2 = 10$)

(A) 02.65

(B) 02.45

(C) 02.75

(D) 02.55

Q.14

A string of length L and radius r elongated by x due to its own weight, when suspended from the ceiling from its one end. If another string of same material, same length and radius double the initial is suspended from its one end, then elongation due to its own weight will be

(A) $4x$

(B) $\frac{x}{2}$

(C) $2x$

(D) x

Q.15

Consider a unit cube which is acted upon by an equal tensile force of F on all the six surfaces of the cube. The increase in the length of each side is fount to be $\frac{F}{Y} - \frac{x\sigma F}{Y}$ where Y, σ are Young's modulus and Poisson's ratio, respectively. Now, Find the value of $x = ?$

(A) 02.15

(B) 02.30

(C) 02.20

(D) 02.00

Answers & Solutions

Q.1 Answer:

3600

Solution:

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$\text{Energy / volume} = \frac{1}{2}\text{Y} \times \text{strain}^2 = 3600\ \text{J m}^{-3}$
$[\text{Strain} = 0.06 \times 10^{-2}]$

Q.2 Answer:

$\frac{1}{2}ky(2x+y)$

Solution:

$W_1 = \frac{1}{2}kx^2$
and $W_2 = \frac{1}{2}k(x+y)^2$
$\therefore \quad W = W_2 - W_1 = \frac{1}{2}k(x+y)^2 - \frac{1}{2}kx^2$
$= \frac{1}{2}ky(2x+y)$

Q.3 Answer:

$\ell/2$

Solution:

$\frac{dF}{dx} = 0$ **at** $\ell/2$

Q.4 Answer:

0.18°

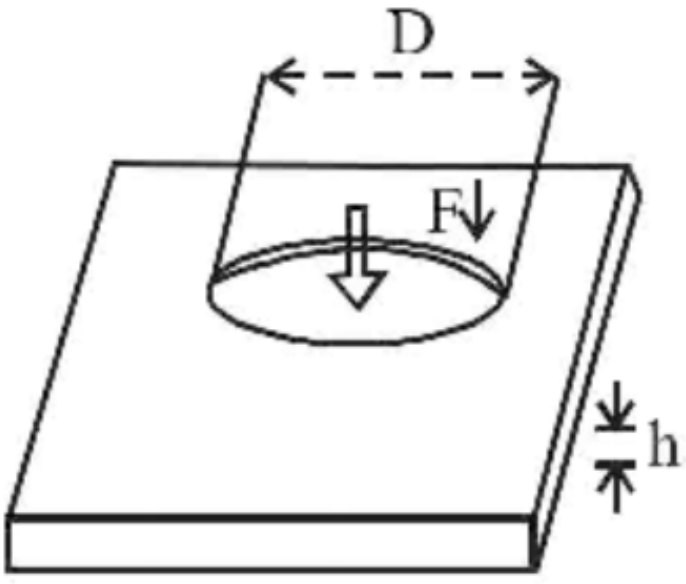
Solution:

$\text{r}\theta = \ell\phi \Rightarrow \phi = \frac{\text{r}\theta}{\ell} = \frac{6\ \text{mm} \times 30^\circ}{\text{lm}} = 0.18^\circ$

Q.5 Answer:

33000

Solution:



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Shearing strain is created along the side surface of the punched disk. Note that the forces exerted on the disk are exerted along the circumference of the disk, and the total force exerted on its center only.

Q.6 Answer:

$\frac{(8.02)^4-(7.98)^4}{(0.8)^4}$

Solution:

$\mathrm{C}_1=\frac{\pi\eta\left(\mathrm{r}_2^4-\mathrm{r}_1^4\right)}{2\ell}, \mathrm{C}_2=\frac{\pi\eta^4}{2\ell}$

Initial volume = Final volume

$\therefore \pi\left[\mathrm{r}_2^2-\mathrm{r}_1^2\right]\ell\rho=\pi\mathrm{r}^2\ell\rho$

$\Rightarrow \mathrm{r}^2=\mathrm{r}_2^2-\mathrm{r}_1^2 \Rightarrow \mathrm{r}^2=\left(\mathrm{r}_2+\mathrm{r}_1\right)\left(\mathrm{r}_2-\mathrm{r}_1\right)$

$\Rightarrow \mathrm{r}^2=(8.02+7.98)(8.02-7.98)$

$\Rightarrow \mathrm{r}^2=16\times0.04=0.64\ \mathrm{cm} \Rightarrow \mathrm{r}=0.8\ \mathrm{cm}$

$\therefore \frac{\mathrm{C}_1}{\mathrm{C}_2}=\frac{\mathrm{r}_2^4-\mathrm{r}_1^4}{\mathrm{r}^4}=\frac{[8.02]^4-[7.98]^4}{[0.8]^4}$

Q.7 Answer:

Shear modulus $<$ Young's modulus $<$ bulk modulus.

Solution:

Poisson's ratio, $\sigma=\frac{\text{lateral strain }(\beta)}{\text{longitudinal strain }(\alpha)}$

For material like copper, $\sigma=0.33$ **And,** $Y=3k(1-2\sigma)$

Also, $\frac{9}{\mathrm{Y}}=\frac{1}{\mathrm{k}}+\frac{3}{\mathrm{n}}$

$\mathrm{Y}=2\mathrm{n}(1+\sigma)$

Hence, $\mathrm{n}<\mathrm{Y}<\mathrm{k}$

Q.8 Answer:

$\mathrm{P}/3\ \mathrm{B}\alpha$

Solution:

Bulk modulus $\mathrm{B}=\frac{-\mathrm{P}}{(\Delta\mathrm{V}/\mathrm{V})}=\frac{-\mathrm{PV}}{\Delta\mathrm{V}}$ **...(1)**

and $\Delta\mathrm{V}=\gamma\mathrm{V}\Delta\mathrm{T}=3\alpha.\mathrm{V}\cdot\mathrm{T}$ **or** $\frac{-\mathrm{V}}{\Delta\mathrm{V}}=\frac{1}{3\alpha.\mathrm{T}} \quad \ldots(2)$

From eqs. (1) and $(2), \mathrm{B}=\mathrm{P}/(3\alpha.\mathrm{T})$ **or** $\mathrm{T}=\frac{\mathrm{P}}{3\alpha\mathrm{B}}$

Q.9 Answer:

$\frac{\Delta V}{V}\propto\frac{1}{B}$

Solution:

$B=\frac{\Delta p}{\Delta V/V}\Rightarrow\frac{1}{B}\propto\frac{\Delta V}{V}[\Delta\mathrm{p}=\text{constant}]$

Q.10 Answer:

$\mathrm{r}_2^4/\mathrm{r}_1^4$

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Solution:

Couple per unit angle of twist, $\mathrm{C}=\frac{\pi \eta \mathrm{r}^4}{2 \ell} \therefore$ **Couple** $\tau=\mathrm{C} \theta=\frac{\pi \eta \mathrm{r}^4 \theta}{2 \ell}$

Here $\eta, \ell, C \& \tau$ **are same. So,** $r^4 \theta=$ **constant** $\therefore \frac{\theta_1}{\theta_2}=\left(\frac{\mathrm{r}_2^4}{\mathrm{r}_1^4}\right)$

Q.11 Answer:

0.5

Solution:

Poisson's ratio σ **is defined as**

$$\sigma=\frac{\text{lateralstrain}}{\text{longitudinal strain}}=\frac{\Delta r / r}{\Delta l / l}$$

where r **is the radius of the wire and** l **its length and** Δr **is the change in** r **and** Δl **the change in** l **when the wire is subjected to tension. Volume of wire before elongation is**

$$V_1=\pi r^2 l$$

Volume of wire after elongation is

$$V_2=\pi(r-\Delta r)^2(l+\Delta l)$$

Given $V_1=V_2$**. Thus**

$$\begin{aligned}\pi r^2 l =& \pi(r-\Delta r)^2(l+\Delta l) \\ =& \pi\left[r^2-2 r(\Delta r)+(\Delta r)^2\right](l+\Delta l) \\ =& \pi r^2(l+\Delta l)-2 \pi r \Delta r(l+\Delta l) \\ & +\pi(\Delta r)^2(l+\Delta l)\end{aligned}$$

Since Δr **and** Δl **are very small, terms of order** $\Delta r \Delta l$ **and** $(\Delta r)^2$ **and higher can be ignored. Then, we have**

$$\pi r^2 l=\pi r^2 l+\pi r^2 \Delta l-2 \pi r l \Delta r$$

$$\text{or } r \Delta l=2 l \Delta r \quad \text{or} \quad \frac{\Delta l}{l}=2 \frac{\Delta r}{r}$$

$$\therefore \quad \sigma=\frac{\Delta r / r}{\Delta l / l}=\frac{1}{2}=0.5,$$

which is choice (d).

Q.12 Answer:

2.5%

Solution:

Poisson's ratio is defined as

$$\sigma=\frac{\Delta d / d}{\Delta l / l} \quad \text{or} \quad \frac{\Delta l}{l}=\frac{\Delta d / d}{\sigma}$$

where d **is the diameter and** l **is the length of the wire. The area of cross-section is**

$$A=\pi r^2=\frac{\pi d^2}{4}$$

$$\text{or} \quad \log A=\log\left(\frac{\pi}{4}\right)+2 \log d$$

Differentiating, we have

$$\frac{\Delta A}{A}=2 \frac{\Delta d}{d}$$

$$\text{or} \quad \frac{\Delta d}{d}=\frac{1}{2} \frac{\Delta A}{A}=\frac{1}{2} \times 2\%=1\%$$

$$\left(\because \frac{\Delta A}{A}=2\%, \text{ given }\right)$$

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Using this in (i) we have

$\frac{\Delta l}{l} = \frac{1\%}{0.4} = 2.5\% \quad (\because \sigma = 0.4)$

Hence the correct choice is (b).

Q.13 Answer:

02.55

Solution:

When the mass is at the lowest point of the vertical circle, the stretching force is

$F = mg + mre^2 = mg + mr\left(2\pi n\right)^2$

$= 15 \times 10 + 15 \times (1) \times (2\pi \times 2)^2 = 2550N$

Elogation $l = \frac{FL}{AY}$

$l = \frac{2550 \times 1}{0.05 \times 10^{-4} \times 2 \times 10^{11}}$

$= 2.55 \times 10^{-3} m$

$\therefore \quad I = 2.55mm$

Q.14 Answer:

x

Solution:

$x = \frac{mg\ell}{2\ AY} = \frac{\rho A\ell g\ell}{2\ Ay}$

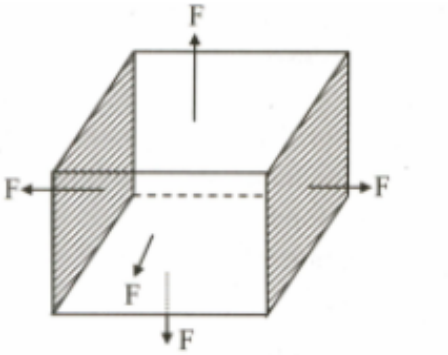
$x = \frac{\rho g\ell^2}{2\ y}$

x

Q.15 Answer:

02.00

Solution:



Tensile strain on each face $= \frac{l}{L} = \frac{FA}{Y}$

For $A = 1 \quad sq \quad Unit$

$\frac{l}{L} = \frac{F}{Y}$

Lateral strain due to force acting on

perpendicular face $= -\sigma \times \frac{l}{L}$

$= -\sigma \times \frac{F}{Y}$

As force is subjected from both directions

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Lateral Strain $= \frac{-2\sigma F}{Y}$

Total increase in length is attributed to both tensile and lateral strain.

Increase in length of each side

$= \frac{F}{Y} + \frac{-2\sigma F}{Y}$

$= \left(1 - 2\sigma\right)\frac{F}{Y}$

$\Rightarrow x = 2$

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